

The Data Detective

Ten Easy Rules to Make Sense of Statistics

Tim Harford

THE ARGUMENT

Statistics can be made to show that storks deliver babies. They also showed that smoking causes lung cancer. How do you take numbers seriously without being fooled by them?

The failure is almost never in the arithmetic. It is in what you wanted the number to say, and in the questions nobody asked before it was published. A claim that flatters you gets waved through; a claim that annoys you gets audited. Reverse that and most of the work is done.

The ten rules are habits of attention rather than techniques. Notice the feeling. Ask what was counted and what the figure should sit beside. Find out who produced it, who was in the sample, and what never got published. Watch the chart as closely as the sentence. Underneath all of them is one instruction: be curious — want to know more about a number than whether it supports you.

THE FRAMEWORK

The Detective's Order of Work

The rules run in a direction: inward first, then at the claim, then at the machinery behind it, then forward to the version of you who will have to admit being wrong. Taken out of order they stop working, because the first one is what makes the rest usable.

01 Notice the feeling

Before evaluating anything, register your own reaction. Outrage, delight and denial all arrive before judgment does, and each of them decides in advance what the evidence will be allowed to mean.

02 Weigh it against your own experience

A statistic and a lived experience are usually answering different questions. Work out which question each one answers instead of asking which of them is lying.

03 Ask what was counted

Every figure rests on a definition somebody chose. Find the definition before arguing about the number, because that is where the real disagreement usually is.

04 Ask what it should sit beside

One number alone is not information. Supply the comparison the presenter left out — over time, or against a total large enough to give the figure a size.

05 Find out where it came from

Who produced it, who was in the sample, who was missed, what never got published, and what the algorithm was actually keying on. The number is the last step of a process, not the first.

06 Keep the case open

Say what you expect in terms clear enough to be shown wrong, and go back and check. The people who forecast best are not the ones who know most; they are the ones willing to change their minds.

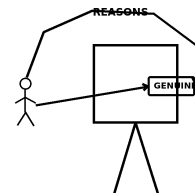
INSIGHTS

What the book claims

INSIGHT 1

Your Feelings Reach the Verdict First

THE FEELING ARRIVES FIRST



Abraham Bredius, the foremost Vermeer scholar of his day, was shown a painting called Christ at Emmaus. He inspected it for every sign of forgery he knew and found none, then declared it genuine, perhaps Vermeer's finest work. He said afterwards that when he saw it, he had difficulty controlling his emotion.

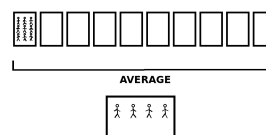
The painting was a forgery, and not even a good one. The remark about his emotion is what explains the failure: the feeling arrived first, and the expert examination that followed was hunting for reasons rather than for the truth. Some studies find that experts are less likely to change their minds when contradicted, not more.

Most numbers carry no charge at all — nobody's pulse rises at the distance to Mars. But a claim touching politics, crime or money produces a reaction before any judgment is made, and from that point evidence is being shopped in rather than weighed. Reading a statistic well starts with noticing what you want it to say.

INSIGHT 2

The Average Is True and the Train Is Full

THE AVERAGE IS TRUE



Presenting a radio programme in London meant a daily commute on a crowded bus and a Tube train bursting at the seams. The published figures said the average bus carried twelve people and the average Tube train fewer than a hundred and thirty. Both came from Transport for London, which counts payment cards tapped at boarding.

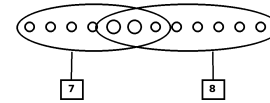
Picture a line running ten trains a day. One carries a thousand passengers and the other nine carry nobody. Average occupancy is a hundred a train, close to London's real figure, and yet every passenger on that line travelled in a crush. The average is accurate and silent about the only thing anyone noticed.

Statistics and personal experience are usually answering different questions, which is why arguing over which one is lying settles so little. On health the statistics win, because they describe the likeliest outcome across the greatest number of people: a chain-smoking grandmother in fine health has not repealed the sixteenfold rise in lung cancer risk.

INSIGHT 3

Every Number Is a Definition in Disguise

WHAT DID YOU COUNT?



In the late 2010s British infant mortality rates varied sharply from one region to another and at first nobody could say why. The answer was a definition. A pregnancy ending at twenty-two or twenty-three weeks was recorded in London as a miscarriage, and in the English Midlands as a live birth followed by an early death.

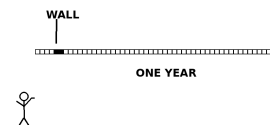
That difference in recording was enough on its own to account for the gap between the two regions' rates. Count the babies who died sounds like the simplest instruction imaginable, and one step in it stops being simple, because the line between a fetus and a baby is genuinely difficult and bitterly contested.

Murky definitions are useful to anyone who wants to move you. A 2017 proposal from a Brexit lobby group would have frozen unskilled immigration for five years, with unskilled meaning anyone earning under thirty-five thousand pounds — most nurses, primary school teachers, paralegals and pharmacists. Support the policy or reject it, but argue about the policy rather than the word.

INSIGHT 4

One Number Alone Is Not Information

ONE NUMBER, NO SCALE



In April 2018 London's newspapers announced that the city's murder rate had overtaken New York's. Fourteen murders in New York that February, fifteen in London. Setting aside that the two cities define murder differently, the claim was technically true — and it supports no conclusion whatsoever.

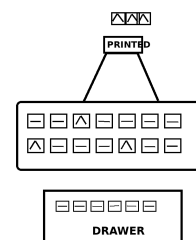
In 1990 London recorded a hundred and eighty-four murders and New York two thousand two hundred and sixty-two. By 2017 the figures were a hundred and thirty and two hundred and ninety-two. Both cities had become far safer, and New York had improved so much that its rate now sometimes dips below London's.

Scale works the way time does. A border wall costing twenty-five billion dollars sounds enormous beside nothing at all; beside a defence budget of just under seven hundred billion a year, about two billion a day, it is a fortnight of military operations. The comparison does not settle the argument. It establishes what size of thing is being argued about.

INSIGHT 5

What Gets Published Is Not What Was Found

WHAT GETS PRINTED



Psychologists Sheena Iyengar and Mark Lepper ran a supermarket stall offering shoppers either twenty-four varieties of jam to taste or six. The larger display drew more people, but three per cent of them bought a jar against thirty per cent at the smaller one. Too much choice paralyses us, ran the conclusion, and it went everywhere.

Gathered together, the published and unpublished research on choice tells a different story. Published papers tended to find a large effect, strongly positive or strongly negative. Unpublished papers tended to find no effect at all. Averaged across the whole body of work, offering more choices came to approximately zero.

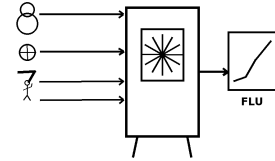
No cheating is required to produce that. Journals prefer surprises, so the studies anyone hears about are a selected sample of the studies that were run; and because careers and incomes depend on publishing, researchers carry a standing incentive to make a finding look more significant than it is. A lone spectacular result is

the least reliable evidence there is.

INSIGHT 6

An Algorithm Finds Patterns, Not Causes

PATTERN, NOT CAUSE



Google Flu Trends counted searches for terms like flu symptoms and pharmacies near me, and estimated new cases faster than the Centers for Disease Control could. Four years after launch it collapsed, announcing a severe outbreak in a winter without one and at one point putting flu at twice the official level.

Google never knew what connected those searches to the disease. The algorithm hunted for correlations and found some with nothing to do with flu, high school basketball among them. What it had built was a general-purpose winter detector, which is also why it missed a summer outbreak in 2009.

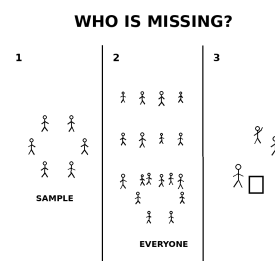
This is not an argument against algorithms. Human judges are neither wholly objective nor consistent in sentencing, and a system comparing a case against similar past ones can be fairer than a person having a bad morning. Telling the two situations apart requires seeing inside, which is what the companies operating these systems are least willing to allow.

ACTIONS

What to do about it

ACTION 1

Ask Who Is Missing From the Sample



Before accepting what a study or a poll found, establish who was in it. Solomon Asch's famous conformity experiments were run on white male American college students in the 1950s, and so much research is run on people who are Western, Educated, Industrialised, Rich and Democratic that the acronym WEIRD was coined for it.

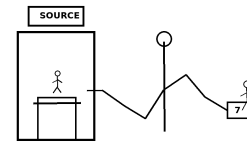
Use this on any finding presented as a fact about people in general, and on every poll. It is working when you can name the population a result actually covers. By 1996 Asch's study had a hundred and thirty-three follow-ups and the effect held; the more diverse ones added that people conform more readily to friends than to strangers.

Polling cannot fix this by recruiting better, because some kinds of people simply respond and others do not — that is sample bias, and no arithmetic afterwards removes it. Where the data was gathered matters too: a survey of American Twitter users describes American Twitter users.

ACTION 2

Trace the Number Back to Whoever Produced It

WHERE DID IT COME FROM?



Follow a figure back to the body that made it, and ask what happens to that body when the number is inconvenient. The Congressional Budget Office, created in 1974, was described by one official as a manhole you drop a bill into and get a cost estimate back from twenty minutes later.

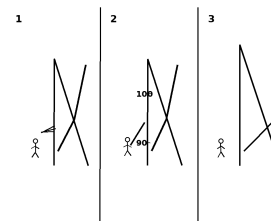
Do this before a number becomes the basis of an argument, and especially when it flatters whoever is quoting it. Jimmy Carter's administration complained that the CBO was not helping when it found his energy efficiency proposals would fall short of their promises. That complaint is the sign of an agency doing its job.

The alternative is on record. Greece, needing to hold its deficit below three per cent of GDP to stay in the eurozone, left several billion euros of borrowing out of its accounts. The European Union found the real figure during the financial crisis, and the Greek economy collapsed.

ACTION 3

Read a Chart in the Same Order You Read a Claim

READ THE AXES



Notice what the picture makes you feel, and only then work out what it says: what the axes are, what is being counted, which dataset or experiment produced it. David McCandless's animation *Debtris* drops costs down the screen as coloured blocks, some of them setting a net figure against a gross one.

Apply it to any chart built to be memorable, which now means most of them. It is working when you can state the graphic's claim

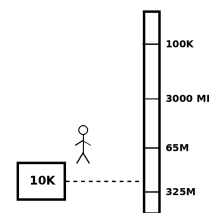
in a plain sentence before deciding whether you believe it. Elegance is not evidence, and a chart that moves beautifully is one you are less likely to interrogate.

Persuasive design is not itself a scandal. Florence Nightingale's rose diagram — two roses side by side, deaths before sanitary measures and after — was built to convince doctors who did not yet know that poor hygiene spread germs. It convinced them, and public health acts followed.

ACTION 4

Keep a Short List of Landmark Numbers

LANDMARK NUMBERS

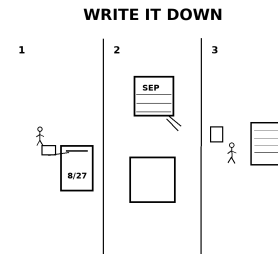


Memorise a handful of reference figures so that a comparison is always to hand without looking anything up — the entrepreneur Andrew Elliott recommends exactly this. The population of the United States is three hundred and twenty-five million, the United Kingdom's is sixty-five million, Boston to Seattle is three thousand miles, and the average novel runs a hundred thousand words.

Reach for the list whenever a figure arrives on its own, which is most of the time. A ten-thousand-word report sounds long until it turns out to be a tenth of a novel. It is working when the comparison is a reflex: something in your head already has something to set beside the new number.

ACTION 5

Write the Prediction Down Where It Can Be Checked



State what you expect in terms clear enough to be declared true or false later, record it, and go back to it. Philip Tetlock collected some twenty-seven thousand five hundred predictions from almost three hundred experts in politics, geopolitics and economics, framed as answerable questions, and then waited eighteen years for the results.

Do it whenever you catch yourself confident about how something will turn out. Tetlock's experts were wrong, overconfident, and misremembered their own forecasts as right where the record showed otherwise. It is working when you hold a record capable of embarrassing you; the alternative is a memory that edits itself in your favour.

Being wrong is not the failure to guard against. Tetlock's second study, drawing on twenty thousand forecasters, found a group who were consistently above average and improving over time. The trait that marked them out was open-mindedness — a willingness to drop a view when the evidence moved.

VOCABULARY

The terms that carry the argument

Each one is a place where a number goes wrong before anybody reads it.

Publication bias

Journals far prefer a surprising result to an inconclusive one, so the findings that reach the public are a selected sample of those produced.

Sample bias

Some kinds of people are likelier to answer a poll than others, so respondents differ systematically from the population. No later calculation removes it.

WEIRD

Western, Educated, Industrialised, Rich and Democratic — the narrow population from which a great deal of psychological research draws its subjects, and therefore its conclusions about people in general.

Big data and algorithms

Big data is the information thrown off by ordinary life — browsing, card payments, phones — collected for reasons unrelated to the question. An algorithm finds patterns in it, reporting correlations it can measure rather than causes it understands.

Average

A single figure standing in for a whole distribution. It can be perfectly accurate and describe nobody's actual experience — ten trains, one full and nine empty, average out to a hundred passengers each.

Statistical bedrock

The official figures a country produces about itself, and the independent agencies producing them. Acting on evidence requires somewhere for evidence to come from.

Landmark numbers

A short memorised list of reference figures — populations, distances, budgets — held in the head so that any new number arrives with something to be compared against.

Superforecaster

One of the minority who predict better than average and improve with practice. What marks them is open-mindedness, not expertise.

QUICK REFERENCE

The ten rules

- 01 Notice how a claim makes you feel before deciding whether it is true.
- 02 Ask which question the statistic answers and which one your experience answers.
- 03 Find out what was counted before arguing about the count.
- 04 Put the figure beside something: a longer timescale, or a bigger total.
- 05 Ask what happened to the studies that found nothing.
- 06 Ask who was in the sample, and who never had the chance to be.
- 07 Judge each algorithm on evidence; demand enough access to judge it.
- 08 Ask who produced the number and whether they answer to the people it flatters.
- 09 Read a chart's axes before accepting what its shape made you feel.
- 10 Say what you expect, record it, and change your mind when it fails.

Be curious. Look deeply for the facts, and keep asking questions.